

DUBAI INTL, United Arab Emirates

WMO: 411940

Lat: 25.255N

Lon: 55.364E

Elev: 10

StdP: 101.20

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	14.8	-0.1	3.8	30.0	2.1	4.4	28.9	9.9	21.0	9.1	20.7	2.2	180	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.2	43.3	23.6	42.1	23.8	41.0	24.1	30.4	35.4	29.8	35.0	29.3	34.8	4.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.1	25.9	33.8	28.7	25.1	33.7	28.0	24.1	33.6	102.4	35.7	99.7	34.9	97.0	35.0	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	7.9	7.2	11.9	46.3	1.4	0.9	11.0	46.9	10.1	47.4	9.4	47.9	8.4	48.5		
			DB	WB	8.1	31.5	1.5	0.5	7.0	31.9	6.2	32.2	5.3	32.5	4.2	32.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.9	20.0	21.4	24.0	28.2	32.3	34.5	36.5	36.7	34.2	30.7	26.0	22.0
	DBStd	6.28	2.01	2.79	3.16	3.26	2.59	2.17	2.18	1.74	1.72	1.77	2.11	2.07
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	9	6	3	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6903	311	319	435	545	692	734	820	827	725	643	481	371
	CDD18.3	3870	58	88	177	295	433	484	562	569	475	384	231	113
	CDH23.3	54513	192	447	1369	3539	6522	7789	9404	9504	7493	5398	2276	578
	CDH26.7	34922	18	114	484	1779	4166	5398	6925	7024	5097	3056	786	73
Wind	WSAvg	3.6	3.4	3.7	3.9	3.7	3.8	3.8	3.9	3.8	3.6	3.3	3.3	3.2
Precipitation	PrecAvg	100	22	17	22	5	0	0	2	0	0	2	7	18
	PrecMax	297	112	100	154	34	1	2	33	3	5	51	41	143
	PrecMin	19	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	74	28	24	36	7	0	0	7	1	1	10	11	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.6	33.0	36.0	39.8	42.6	44.1	45.1	45.1	42.2	38.9	34.2	30.1
		MCWB	17.7	17.9	18.3	19.9	21.0	22.5	23.6	23.2	22.7	21.4	19.3	18.7
	2%	DB	26.9	29.8	33.2	37.2	40.8	42.0	43.4	43.1	40.8	37.1	32.8	28.2
		MCWB	17.2	17.3	18.2	19.6	21.4	23.1	24.1	23.9	23.4	22.0	20.7	19.3
	5%	DB	25.2	27.8	31.1	35.8	39.1	40.5	42.0	41.9	39.1	35.9	31.4	27.2
		MCWB	17.6	17.5	18.2	19.5	21.4	23.7	24.7	24.4	23.9	22.7	21.2	19.4
	10%	DB	24.1	26.0	29.2	34.0	37.8	39.0	40.5	40.4	38.0	34.9	30.2	26.2
		MCWB	17.4	17.5	18.2	19.7	21.9	24.3	25.2	25.1	24.7	23.4	21.4	19.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	21.3	22.6	24.7	28.0	30.1	31.1	31.2	30.2	28.9	25.4	22.7
		MCDB	24.4	25.3	27.4	30.8	33.7	35.3	36.2	35.9	34.5	33.2	30.0	26.4
	2%	WB	19.8	20.5	21.7	23.7	26.9	29.3	30.4	30.4	29.7	28.0	24.2	21.8
		MCDB	23.9	24.7	26.4	30.1	32.9	34.6	35.7	35.4	34.1	32.3	29.3	25.9
	5%	WB	19.1	19.7	21.1	22.9	26.0	28.8	29.8	29.8	29.1	27.2	23.5	21.0
		MCDB	23.4	24.2	25.9	29.6	32.6	34.2	35.3	35.1	33.9	31.9	28.8	25.3
	10%	WB	18.4	19.1	20.5	22.1	25.2	28.1	29.2	29.2	28.6	26.5	22.7	20.3
		MCDB	22.8	23.8	25.5	28.8	32.5	34.0	35.2	35.1	33.6	31.6	28.2	24.9
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.5	8.9	10.2	10.9	10.3	9.0	9.2	9.3	9.4	8.6	8.2
		MCDBR	9.6	10.6	11.5	12.2	12.3	11.7	10.4	10.5	10.3	10.3	9.6	9.3
		MCWBR	4.6	4.9	5.6	5.8	6.0	6.6	6.4	6.4	6.6	6.2	4.8	4.6
	5% WB	MCDBR	8.4	8.7	9.1	10.4	11.2	10.3	8.8	9.1	9.2	9.2	8.9	8.5
		MCWBR	4.7	4.8	4.9	5.6	6.1	5.9	5.6	5.8	5.7	5.3	4.7	4.7
Clear-Sky Solar Irradiance	taub	0.443	0.502	0.563	0.595	0.588	0.637	0.726	0.689	0.599	0.514	0.457	0.438	
	taud	2.088	1.918	1.774	1.731	1.742	1.687	1.569	1.638	1.773	1.934	2.084	2.105	
	Ebn at Noon	795	773	749	737	740	700	639	661	711	751	774	780	
	Edh at Noon	144	183	221	236	233	245	275	256	219	177	143	136	
All-Sky Solar Radiation	RadAvg	4.05	4.84	5.72	6.60	7.41	7.55	7.11	6.74	6.24	5.43	4.38	3.83	
	RadStd	0.26	0.38	0.39	0.40	0.16	0.14	0.19	0.15	0.09	0.08	0.13	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	+0.90	+1.31	N/A	+0.55	N/A	N/A	N/A	-5	+346	+343	
Regional (15 neighbors)	+0.47	+0.70	N/A	N/A	N/A	N/A	N/A	+150	+149		

Nomenclature: See separate page